

5 FLIP FLOP

5.1 Theoretical part

5.1.1 Sequential circuits:

Sequential circuits consist of a combinational circuit and memory elements, as in figure 1.

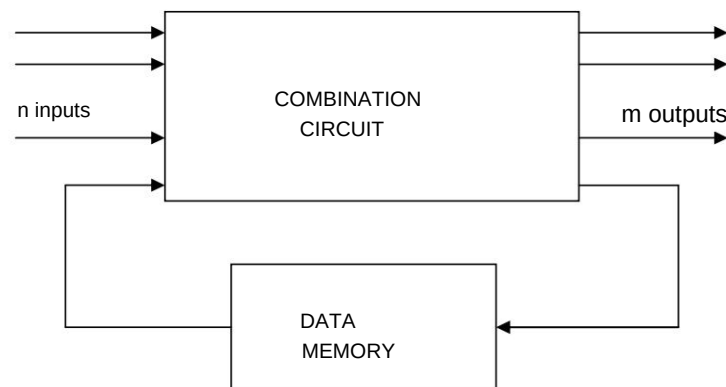


Fig.1: Block diagram of a sequential circuit Fig. 1
Block diagram of a sequential circuit

The memory elements store binary information that constitutes the current state of the sequential circuit at any given time. The next state of the outputs of a sequential circuit is therefore a function of its inputs and the current state of its outputs. There are two categories of sequential circuits:

1. synchronous sequential circuits
circuits)
2. asynchronous sequential circuits
circuits)

In an asynchronous sequential circuit, the memory elements consist of logic gates (which introduce a propagation delay in the signals propagating through them) and are called latches. Some of the outputs of the combinational circuit contained in a sequential circuit are connected to the memory elements, whose outputs feed some inputs of the combinational circuit (feedback loop). In a synchronous sequential circuit, the memory elements are flip-flops. The flip-flop is used as a memory

cell because it is a circuit that can be maintained in one state (storing 1 bit) until an appropriate input signal causes it to change state. In a synchronous sequential circuit, a master clock generator

generator) supplies the circuit with clock pulses that are distributed throughout the circuit to achieve synchronization. Synchronous sequential circuits where clock pulses are applied to the memory elements (flip-flops) are called clocked sequential circuits.

5.1.2 Latches

The latch has two inputs: - S (Set)

- R (Reset) and two

outputs: - Q

output

- $\bar{Q}$ complement of the output

The basic circuit of a latch can be implemented with two NAND gates or two NOR gates where the output of each gate is cross-connected to the input of the other gate creating a feedback loop. This type of latch is also called a direct-coupled RS flip-flop or SR latch.

5.1.3 Latch with NAND gates

The latch circuit can be implemented with two gates NAND as in figure 2.

Fig. 2 Latch with NAND gates

The NAND gate latch is an asynchronous sequential circuit consisting of:

- two NAND gates and has: -

two inputs R (Reset) and S

(Set) - two outputs Q and $\bar{Q}$ The

state of the latch is the

value of the output Q. The (useful) states in which it can be found are: -

the set state where $Q = 1$ and $\bar{Q} = 0$ - the reset

or clear state where $Q = 0$

and $\bar{Q} = 1$

The operation of the NAND gate latch is described below:

1. $S=1$ and $R=1$

Resting state of the latch. This is the steady state of the latch because the output remains unchanged (the outputs retain their values that were previously set at inputs $S=1$ and $R=1$).

2. $S=0$ and $R=1$

Activating the latch. The output becomes $Q=1$ (position) and remains $Q=1$.

3. $S=1$ and $R=0$

Resetting the latch. The output becomes $Q=0$ (resetting) and remains $Q=0$.

4. $S=0$ and $R=0$

Unused state of the latch. The outputs are $Q=1$ and $\bar{Q}=1$. This situation should not be used.

If the latch is in the set state ($S=0$ and $R=1$ with $Q=1$) and if $S=1$ is applied, then the flip-flop remains in the set state ($Q=1$) while if the flip-flop is in the reset state ($S=1$ and $R=0$ with $Q=0$) and if $R=1$ is applied, then the flip-flop remains in the zero state ($Q=0$). In the two above cases the outputs remain unchanged (the locksmith has a memory).

When the status of the locksmith is about to change, then:

If the latch is in the set state ($S=0$ and $R=1$ with $Q=1$), then we first make $S=1$ so the latch goes to rest, and then $R=0$, so the latch goes to the zeroing state ($Q=0$). If the latch is in the reset state ($S=1$ and $R=0$ with $Q=0$), then First we set $R=1$ so the latch goes to rest and then $S=0$, so the latch goes to position state ($Q=1$).

The latch can be found in an unused state (where $Q=1$ and $\bar{Q}=1$) if $S=0$ and $R=0$ simultaneously (since $S=1$ and $R=1$). We will prevent the locksmith from finding himself in this situation. The locksmith must be prevented from finding himself in this situation.

Table 1 shows the truth table of the latch with NAND gates, which summarizes its operation.

S	R	Q	$\bar{Q}$	Observations
1	0	1	0	Position
1	0	0	1	Zeroing
1	1	1	1	Unused
1	1	0	0	Unused

The output Q follows the input R when $S=1$

Table 1 Truth table of a latch with NAND gates

5.1.4 Latch with NOR gates

The latch circuit can be implemented with two gates NOR as shown in figure 3.

Fig. 3 Latch with NOR gates

The NOR gate latch is an asynchronous sequential circuit consisting of: - two (2)

NOR gates and has: - two (2)

inputs R (Reset) and S (Set) and - two (2)

outputs Q and $\bar{Q}$

The state of the latch is the value of the output Q. The useful situations in which it can be found are the following:

position status (set)

where $Q = 1$ and $\bar{Q} = 0$

reset or clear status

where $Q = 0$ and $\bar{Q} = 1$

Describe the operation of the NOR gate latch with analog in a manner similar to that of the operation of a latch with NAND gates, which mentioned previously. It is noted that the resting state of latch with NOR gates is for $R=0$ and $S=0$.

Table 2 shows the truth table of the mandolin with NOR gates, where its operation is summarized.

S	R	Q	$\bar{Q}$	Observations
0	0	Q	$\bar{Q}$	Unchanged
0	1	0	1	Zeroing
1	0	1	0	Position
1	1	0	0	Not used

The output Q follows the input S when SR

Table 2 Truth table of a latch with NOR gates

5.1.5 Flip-Flop

The flip-flop is a synchronous sequential circuit, its outputs which respond to the inputs when clock pulses are applied

(clock pulses) to one of its inputs called the clock input (CP). The most commonly used types of flip-flops are the following:

RS flip-flop	JK flip flop
D flip-flop	T flip-flop

In explaining the operation of flip-flops, we consider the current state of the output Q as $Q(n)$ and the next state as $Q(n+1)$. The next state always occurs after the application of a clock pulse.

5.1.6 RS Flip-Flop

The RS flip-flop can be implemented with four NAND gates, as shown in Figure 4. The operation of the RS flip-flop is described below:

1. When $S=0$ and $R=0$, then the next state (output Q) is same as the previous situation.
2. When $S=0$ and $R=1$, then the next state is $Q=0$ (Reset)
3. When $S=1$ and $R=0$, then the next state is $Q=1$ (Set)
4. When $S=1$ and $R=1$, then the next state is undefined. This is an unused state.

Fig.4 RS flip-flop

The characteristic table of a flip-flop gives us the next (after a clock pulse) state of the output of $Q(n+1)$ in relation to the values of its inputs and the current state $Q(n)$. The characteristic table of the RS flip-flop is shown below (table 3).

SR	Q(n)	0 0 0 0	0 0 1 0 1	Q(n+1)
1	0	0	1	0
				1
			1	0
				0
			0 1 0	1
1			1	1
1			0	Unspecified
1	1		1	Unspecified

and in summary:

SR	0 0 0	Q(n+1)
1 1 0		Q(n)
		0
		1
1	1	Unspecified

*unused condition

Table 3 RS flip-flop characteristic table

The flip-flop excitation table shows how to transition from present state to the next state and results from the characteristic table of each flip-flop. The excitation table of the RS flip-flop is shown in Table 4.

s	R	Q(n)	Q(n+1)
0	X		0
	0	0 0	1
1 0	1	1	0
X	0	1	1

Table 4 RS flip-flop excitation table

The characteristic equation of a flip-flop gives the following state of the output $Q(n+1)$ with respect to its inputs and the current state $Q(n)$ algebraically. The characteristic equation results from the characteristic table after simplification with Karnaugh table.

For the RS flip-flop (with the SR term equivalent to $SR=0$) we have:

$$Q(n+1) = S + \bar{R} Q(n)$$

The symbol of RS flip-flop is shown in figure 5.

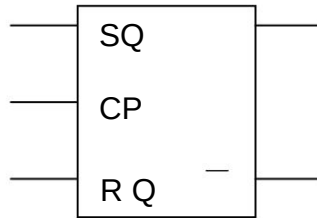


Fig. 5 Symbol of RS flip-flop

5.1.7 D Flip-Flop

Eliminating the unwanted behavior and undefinedness in the unused state of the RS flip-flop is achieved with the D flip-flop. The D flip-flop can be implemented using an RS flip-flop and a NOT gate, as shown in Figure 6.

Fig. 6 D flip-flop

The operation of the D flip-flop is described below:

When the clock pulse input CP is 0, then the flip-flop cannot change state, regardless of the value of the D input because the RS flip-flop does not change state. When the clock pulse input CP is 1, then if D=0 we have Q=0 because S=0 and R=1 while if D=1, then we have Q=1 because S=1 and R=0. The name of the D flip-flop comes from

its ability to store data (Data flip-flop) and delay its propagation

(Delay flip-flop). The information at the data input D of the D flip-flop is transferred to the output Q of the flip-flop when the CP becomes 1. The characteristic table of the D flip-flop is shown in Table 5.

DQ(n)	0 0 0 1 1	Q(n+1)
0 1 1		0
		0
		1
		1

and in summary:

D	Q(n+1)
0	0
1	1

Table 5 D flip-flop characteristic table

The characteristic equation of the D flip-flop gives us:

$$Q(n+1) = D$$

The symbol of the D flip-flop is shown in figure 7.

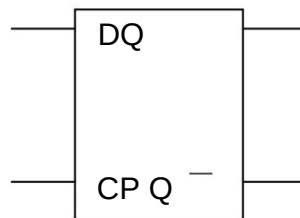


Fig. 7: Symbol of D flip-flop
Fig. 7 Symbol of D flip-flop

5.1.8 JK Flip-Flop

The unused state of the RS flip-flop ($S=1$ and $R=1$) is also avoided by using the JK flip-flop. In the JK flip-flop when the inputs J (set) and K (reset) are set to $J=1$ and $K=1$, then the flip-flop changes state (i.e. if the output was $Q=0$ then it will become $Q=1$ and vice versa).

The JK flip-flop can be implemented with four NAND gates as in figure 8

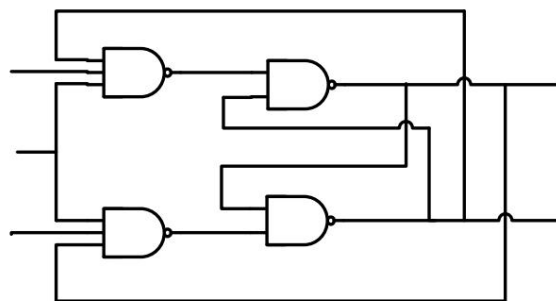


Fig. 8 JK flip-flop

The operation of the JK flip-flop is as follows:

1. When $J=0$ and $K=0$, then the next situation is the same as previous situation.
2. When $J=0$ and $K=1$, then the next state is $Q=0$.
3. When $J=1$ and $K=0$, then the next state is $Q=1$.
4. When $J=1$ and $K=1$, then the state of the flip-flop is reversed, that is, the next state is complementary to the previous situation.

The characteristic table of the JK flip-flop is shown in table 6.

JK Q(n) Q(n+1)			
0 0 0 0	0 1		0
			1
0	1	0	0
0	1	1	0
	0	0	1
1 1	0		1
1	1	1	1
1	1	0 1	0

and in summary:

JK Q(n+1)		
0	0	$Q(n)$
0 0		
1	1	1
1	0 1	$\overline{Q(n)}$

Table 6 JK flip-flop characteristic table

The excitation table of the JK flip-flop is shown in Table 7.

J	KQ(n) Q(n+1)		
0	X 0		0
1	X 0		1
X	1	1	0
X	0	1	1

Table 7 JK flip-flop Excitation Table

The characteristic equation of the JK flip-flop is the following:

$$Q(n+1) = JQ(n) + K\overline{Q(n)}$$

The symbol of the JK flip-flop is shown in Figure 9.

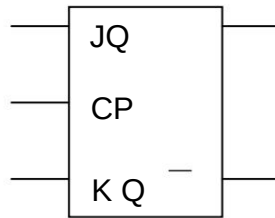


Fig.9: Graphical symbol of JK flip-flop
Fig. 9 Symbol of JK flip-flop

5.1.9 T Flip-Flop

If the J, K inputs of the JK flip-flop are connected together, then a variant of the JK flip-flop, the T flip-flop, is obtained. The T flip-flop has one input, T, and can be implemented as in Figure 10.

Fig. 10 T flip-flop

The operation of the T flip-flop is described below:

1. When $T=0$, then the next state is the same as the previous state.
2. When $T=1$, then the state of the flip-flop is inverted, that is, the next state is the complement of the previous state.

The name of the T flip-flop comes from its ability to invert (Toggle) its state. The characteristic table of the T flip-flop is shown in Table 8.

T	Q(n)	Q(n+1)
0	0	0
0	1	1
	0	1
1 1	1	0

and in summary:

T	Q(n+1)
0	Q (n)
1	$\overline{Q}_{(n)}$

Table 8 T flip-flop characteristic table

The characteristic equation of the T flip-flop is the following:

$$Q(n+1) = T \overline{Q(n)} + \overline{T} Q(n)$$

The symbol of the T flip-flop is shown in figure 11.

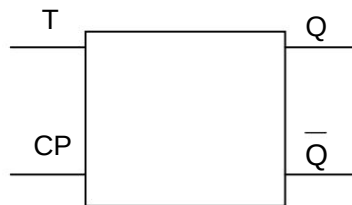


Fig. 11 Symbol of T flip-flop

5.1.10 Master-slave Flip-Flop

In all the flip-flops we have examined so far, the time t_{on} should be the clock pulse is at logic 1, be less than the total signal propagation delay time from the input to the output t_{pd} circuit so that new output states when they appear do not feed back to the input gates as long as the clock pulse lasts. Otherwise the flip-flop will become unstable, changing continuously the states of its outputs until the clock input becomes 0. Also, the period of the clock pulse T should be such that the next clock pulse to occur after the new value has stabilized expense status.

So for correct operation it must be true:

$$t_{on} < t_{pd} < T$$

If the above relationship does not hold, then for example in the JK flip-flop when $J=1$ and $K=1$ and the clock pulses ($CP=1$) have a long duration, then the state of the flip-flop, after being inverted once, is inverted continuously throughout the duration of the clock pulse (i.e. until $CP=0$). Generally in such a case the truth table is not followed and the output of the flip-flop is undefined.

The condition $t_{on} < t_{pd}$ is not always easy to satisfy because the time t_{pd} is very short as we have already seen. To address this problem, Master-Slave flip-flops were invented. Figure 12 below shows the circuit of a Master-Slave JK flip-flop.

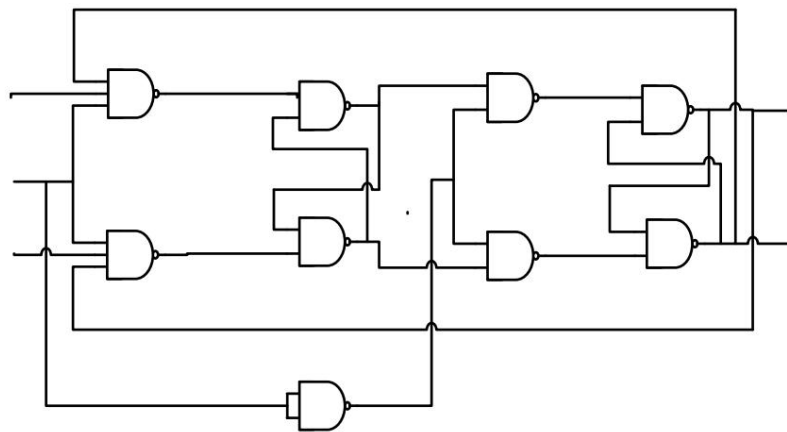


Fig. 12 Master-Slave JK flip-flops

As shown in Figure 12, the first flip-flop called the master is activated with the clock pulse at logic 1 while the second flip-flop called the slave is activated with the clock pulse at logic 0. Thus, the state change appears on the slave outputs after the clock pulse becomes logic 0, and the new states are fed back to the master flip-flop when the clock pulse is already at logic 0, resulting in no new changes in the outputs of the master and therefore of the slave, which are also the outputs of the entire Master-Slave flip-flop.

5.1.11 Excitation of Flip-Flops

The state of a latch or flip-flop is changed by a change in an input signal called triggering. Latches are triggered by a change in the value (logic level) of the signals at their inputs, while flip-flops are triggered by their clock pulses, which can be positive or negative. A source of positive clock pulses remains at "0" during the interval between pulses and goes to "1" during the pulse. Therefore, there are two signal transitions for each pulse:

- the transition from "0" to "1" which is called a positive transition (Positive Going Transition - PGT) or a rising transition or a positive edge or a positive front. - the transition from "1" to "0"

which is called a negative transition (Negative Going Transition - NGT) or a falling transition or a negative edge or a negative front.

The two transitions of the clock pulses are shown in Figure 13.



Fig. 13 Clock pulse transitions

Flip-flop state changes occur either during the clock pulse being 1 or 0 (level triggering) or with the positive or negative edge of the clock pulse (edge triggering).

Edge-triggered flip-flop symbols are shown in Figure 14. The triangle at the clock input indicates that the flip-flop is triggered by the positive edge of the clock, while the circle with the triangle indicates that the flip-flop is triggered by the negative edge of the clock.

Fig. 14 Symbols of edge-triggered flip-flops

5.1.12 Asynchronous inputs

The S, R, J, K, D, and T inputs of the flip-flops discussed earlier are called synchronous inputs because their effect on the flip-flop outputs is synchronized with the CP input of the clock pulse. However, many flip-flop integrated circuits have two additional

inputs called asynchronous inputs because their effect on the flip-flop outputs does not depend on the clock pulses.

Asynchronous inputs determine the state of the flip-flop independently of the values of its synchronous inputs and the clock and are commonly used to set the flip-flop to a certain initial state (position or reset) before starting to operate with the clock. Asynchronous inputs are:

The preset (PRESET) used to set the flip-flop to the position state $Q=1$ and the clear (CLEAR) used to set the flip-flop to the reset state $Q=0$.

For example, Figure 15 shows the symbol for a JK flip-flop with asynchronous inputs.

Fig. 15 JK flip-flop symbol with asynchronous inputs

The operation of the above JK flip-flop with asynchronous inputs is described below: When

PRESET=1 and CLEAR=1 then the asynchronous inputs are deactivated (the circles on the asynchronous inputs mean that they are activated with a logical state 0 – active low) and the outputs of the flip-flop respond to the synchronous inputs J and K as well as to the pulses of the CP clock, that is, the timing function is performed. When PRESET=0 and CLEAR=1 then the flip-flop is set to the position state $Q=1$ When PRESET=1 and CLEAR=0, then the flip-flop is set to the

reset state $Q=0$ PRESET=0 and CLEAR=0 must not be simultaneously PRESET=0 and

CLEAR=0 Then, in table 9, the complete operation table of such a JK flip-flop and specifically the 74LS76 as presented in the data sheets of the integrated circuit manufacturers is given.

PR	CLR	CLK	J	K	Q	$\overline{Q}$
L	H	X	X	H	L	
H	L	X	X	L	H	
L		X	X	X	H*	H*
H	H				0	0
H	H			0	1	0
H	H		1		0	1
H	H		0	1	TOGGLE	

Notes:

The symbol indicates that the flip-flop is triggered by the negative edge of the clock pulse.

* Outputs do not remain stable when asynchronous inputs PRESET and CLEAR be deactivated and become logical 1.

Q0 is the previous state of the flip-flop (before the application of the new inputs J and K and of course the clock pulse).

TOGGLE: the state of the flip-flop is inverted, i.e. the next state is complementary to the previous one

Table 9 Complete functional description of the 74LS76 JK flip-flop

5.2 Work Execution

1) Connect the latch circuit with NAND gates (figure 16) and prove its truth table (table 1).

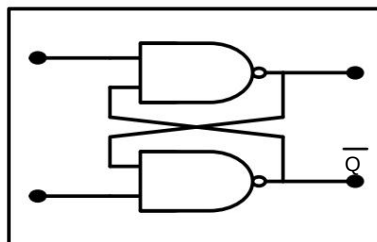


Fig. 16 Latch with NAND gates.

2) Connect the RS Flip-Flop circuit (figure 17) and verify the table attribute (table 3).

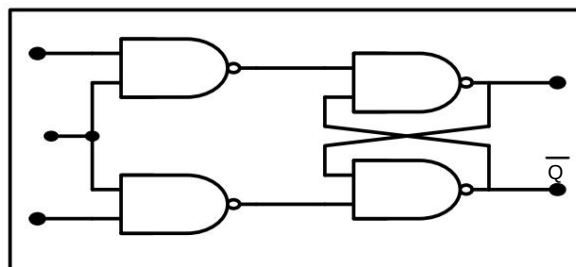


Fig. 17 RS Flip-Flop

The CP will be connected to the active low button of the test bench where by pressing it momentarily we will give a clock pulse after we have given the values we want to the R, S inputs. Do not disconnect the circuit.

3) By adding an inverter between the S and R inputs of the previous circuit of figure 17, create a D Flip-Flop (figure 18) and verify its characteristic table (table 5).

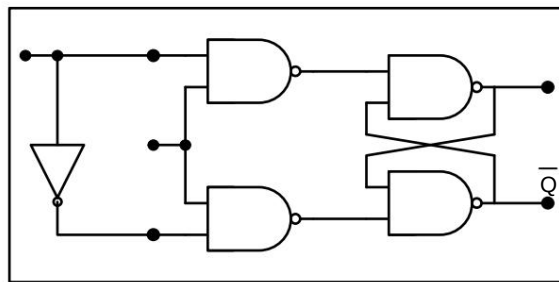


Fig. 18 Conversion of RS Flip-Flop to D Flip-Flop.

4) Using the integrated 74LS76 which is a dual JK Flip-Flop whose pinout is given in figure 19, connect the asynchronous and synchronous inputs PR1, CLR1, J1, K1 to switches while the clock input CLK1 to button active low and the outputs Q1 and Q 1 to LEDs. Verify table 9 which gives the complete description of the 74LS76 JK Flip-Flop.

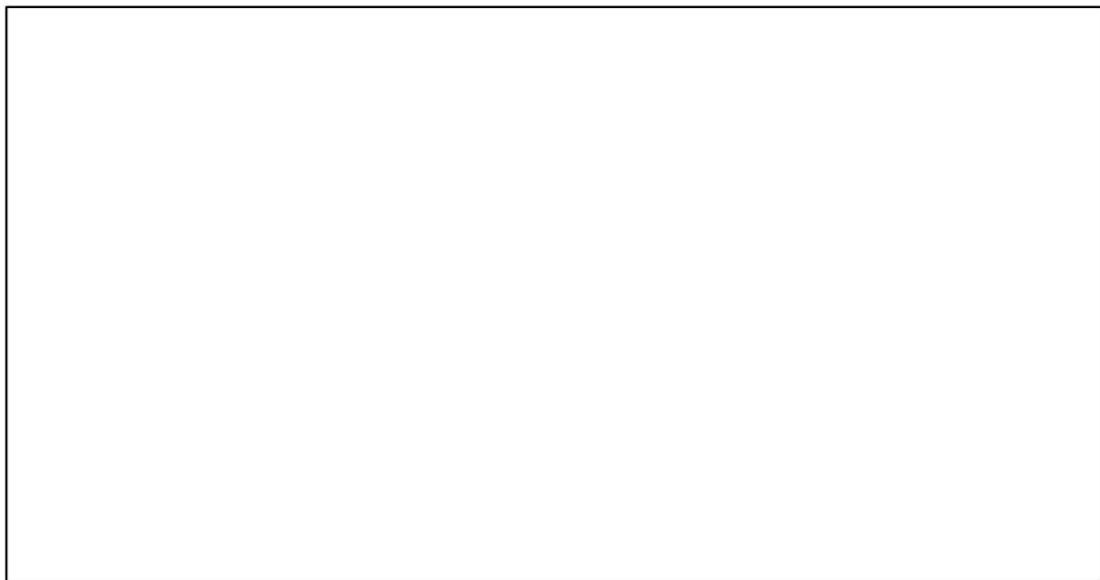


Fig. 19 Function tables and terminal diagram of the 7476 and 74LS76 integrated circuits

5.3 Questions

1. Why do you think modern sequentials are needed?
circuits? 2.

When does the next state appear in a Flip-Flop? 3. What is the difference between the SR latch and the SR Flip-Flop? 4. Where does the name D Flip-Flop come from? 5. Design a T Flip-Flop based on the JK Flip-Flop. Write its characteristic operation table. 6. What is the condition for proper operation of Flip-Flops, and why were Master-Slave Flip-Flops designed: 7. The following D Flip-Flops (figure 20) have initial states

$Q_0=Q_1=0$. Give the outputs time correspondence with the clock Q_0 and Q_1 until a complete cycle of operation of the circuit is seen.

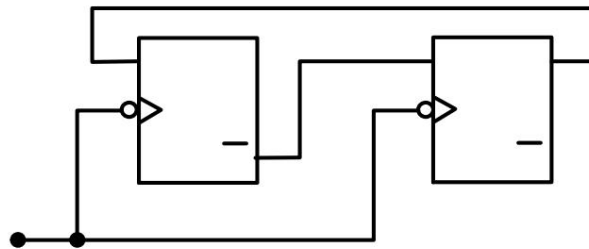


Fig. 20 Question circuit 7